

Digital Float Control System - Implementation Status

Table of Contents

1. Introduction	1
2. System Overview	2
3. Bootloader Requirements	2
4. Application Requirements	3
5. Hardware Interface Requirements	4
6. User Configuration Requirements	5
7. Safety and Diagnostics Requirements	6
8. Protocol Requirements	6
9. Proposed New Features	7
9.1. Security Enhancements	7
9.2. Reliability Improvements	7
9.3. Diagnostics and Monitoring	8
9.4. User Interface Enhancements	8
9.5. Functional Extensions	8
9.6. Performance Optimizations	9

1. Introduction

This document provides a detailed overview of the implementation status for the Digital Float Control System as of 2025-04-15

Each requirement is categorized and includes both a progress percentage and color-coded status indicator:

- 85-100% - High completion (Green)
- 60-84% - Medium completion (Orange)
- 0-59% - Low completion (Red)

Summary of Progress

Category	Average Completion
System Overview	78%
Bootloader Requirements	71%
Application Requirements	76%

Category	Average Completion
Hardware Interface Requirements	80%
User Configuration Requirements	83%
Safety and Diagnostics Requirements	64%
Protocol Requirements	83%

2. System Overview

Requirement	Status	Completion
The system shall control aircraft landing gear and rudder motors	Medium	[85%] - Core functionality implemented, but edge cases and failure modes need further testing
The system shall provide visual status indication via LEDs	High	[90%] - LED control implemented, but could benefit from improved user feedback during critical states
The system shall implement a secure bootloader for firmware updates	Medium	[60%] - Basic bootloader implemented, lacks signature verification and rollback protection

3. Bootloader Requirements

Requirement	Status	Completion
The bootloader shall support in-field firmware updates	Medium	[75%] - Basic update functionality implemented, but lacks robust error recovery and verification

Requirement	Status	Completion
The bootloader shall verify application integrity before execution	Low	[40%] - Only checks for valid reset handler, no CRC/hash verification or signature validation
The bootloader shall provide version information on request	High	[90%] - Version reporting implemented, but format could be enhanced with more metadata
The bootloader shall have a configurable wait time before jumping to application	Medium	[80%] - Hard-coded constant implemented, but lacks runtime configuration options

4. Application Requirements

Requirement	Status	Completion
The application shall control up to 6 separate control channels	High	[90%] - Channels initialized, but concurrent operation edge cases need verification
The application shall support both landing gear and rudder control types	High	[85%] - Basic differentiation implemented, but state transitions could be more robust
The application shall monitor voltage and current for each channel	Medium	[70%] - Basic monitoring implemented, but calibration and accuracy verification needed

Requirement	Status	Completion
The application shall detect and report error conditions	Low	[50%] - Limited error detection, lacks comprehensive error classification and reporting
The application shall provide configurable LED color indication based on system state	High	[85%] - Color states implemented, but user feedback during transitions could be improved
The application shall store and recall user settings from non-volatile memory	Medium	[75%] - Basic storage implemented, but lacks data validation and corruption recovery
The application shall store and recall channel configuration from non-volatile memory	Medium	[75%] - Basic storage implemented, but needs data integrity checks

5. Hardware Interface Requirements

Requirement	Status	Completion
The system shall interface with I2C power monitoring devices	Medium	[80%] - Basic I2C communication implemented, but bus error handling needs improvement
The system shall interface with I2C relay control devices	Medium	[80%] - Basic control implemented, but retry mechanisms and failure detection need enhancement

Requirement	Status	Completion
The system shall read switch inputs for landing gear and rudder position control	High	[85%] - Switch reading implemented, but debouncing and failure detection could be improved
The system shall support a test switch for diagnostics and configuration	Medium	[80%] - Basic test functionality implemented, but interface could be more intuitive
The system shall communicate via UART for commands and diagnostics	Medium	[75%] - Basic UART communication implemented, but lacks robust error handling and flow control

6. User Configuration Requirements

Requirement	Status	Completion
The system shall support adjustable LED brightness	High	[85%] - Brightness adjustment implemented, but limited range and granularity
The system shall support customizable LED colors for different states	Medium	[80%] - Color customization implemented, but UI for configuration is limited



The customizable LED colors include support for:

- Landing gear up position
- Landing gear down position
- Rudder up position
- Rudder down position
- Rudder inactive position

- Moving/transition states
- Error states

7. Safety and Diagnostics Requirements

Requirement	Status	Completion
The system shall provide hardware relay self-test capability	Medium	[70%] - Basic test implemented, but comprehensive failure detection and reporting needed
The system shall monitor power parameters for error detection	Medium	[60%] - Basic monitoring implemented, but lacks clear thresholds and response policies
The system shall provide I2C device scanning for diagnostics	Medium	[75%] - Basic scanning implemented, but results interpretation and remediation guidance lacking
The system shall implement watchdog/reset capability	Medium	[60%] - Reset command implemented, but watchdog integration not evident
The system shall have clear error indication for fault conditions	Low	[55%] - Basic error indication, but lacks detailed error codes and recovery guidance

8. Protocol Requirements

Requirement	Status	Completion
The system shall implement a command-based communication protocol	Medium	[80%] - Protocol framework implemented, but lacks authentication and error recovery
The system shall support commands for system functions	High	[85%] - Basic commands implemented, but some need extended functionality



The system supports commands for:

- Version reporting
- Device reset
- Hardware diagnostics
- Configuration updates
- Status monitoring
- Test mode activation

9. Proposed New Features

9.1. Security Enhancements

Feature	Priority	Benefit
Implement cryptographic signature verification for firmware updates	High	Prevents installation of unauthorized firmware
Add secure bootloader with rollback protection	High	Prevents downgrade attacks and installation of vulnerable firmware
Implement authentication for protocol commands	Medium	Prevents unauthorized access to configuration and control functions

9.2. Reliability Improvements

Feature	Priority	Benefit
Implement CRC validation for all stored settings	High	Prevents corruption-based failures
Add redundant storage for critical configuration data	Medium	Improves resilience against flash memory failures
Implement comprehensive watchdog system	High	Ensures system recovery from software hangs or crashes

9.3. Diagnostics and Monitoring

Feature	Priority	Benefit
Create detailed error logging system with persistent storage	Medium	Enables post-mortem analysis of failures
Implement comprehensive built-in self-test (BIST) functionality	Medium	Allows detection of hardware issues during initialization
Add power consumption profiling and anomaly detection	Low	Enables predictive maintenance and failure prediction

9.4. User Interface Enhancements

Feature	Priority	Benefit
Develop more intuitive configuration interface for LED patterns	Low	Improves user experience during configuration
Add support for piezo buzzer for audible feedback	Medium	Provides alternative notification method for critical states
Implement customizable blinking patterns for error conditions	Low	Allows better differentiation between error types

9.5. Functional Extensions

Feature	Priority	Benefit
Implement position feedback sensing for more accurate state tracking	Medium	Improves accuracy of position reporting
Add support for ambient light sensing to auto-adjust LED brightness	Low	Improves visibility in varying lighting conditions
Implement motor current profiling for mechanical issue detection	Medium	Enables early detection of mechanical problems

9.6. Performance Optimizations

Feature	Priority	Benefit
Optimize power consumption during idle states	Medium	Reduces system power requirements

Report generated on 2025-04-15

```
<style>
@page {
  @bottom-center {
    content: "Page " counter(page) " of " counter(pages);
    font-size: 9pt;
  }
  @bottom-left {
    content: "© 2025 Skymatik Aero";
    font-size: 9pt;
  }
  @bottom-right {
    content: "Implementation Status v1.0";
    font-size: 9pt;
  }
}
</style>
```